

DAY 02: AWS EC2 & RELATED CORE CONCEPTS

Cloud Computing & DevOps Master Curriculum

1. Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 provides resizable compute capacity in the Amazon Web Services (AWS) cloud. It allows users to launch virtual servers called **EC2 Instances** on demand without upfront hardware investments.

Key benefits include scaling capacity up or down within minutes, pay-per-second billing, and full administrative root access.

2. Core Components of EC2

Component	Description
AMI (Amazon Machine Image)	A pre-configured template containing OS, application server, and software stack required to launch an instance.
Instance Types	Hardware configurations categorized by vCPU, memory, storage, and networking performance (e.g., t3.micro, c6g.large, r5.xlarge).
User Data	Shell scripts executed automatically on instance boot up for bootstrapping (installing updates, docker, nginx).
Elastic IP (EIP)	A static, public IPv4 address designed for dynamic cloud computing that remains attached to your instance across restarts.

3. EC2 Instance Naming & Classification

EC2 instances follow a standardized naming convention: **[Family][Generation].[Size]** (e.g., **t3.micro**).

- **General Purpose (t3, m5, m6i):** Balanced compute, memory, and networking for web servers and dev environments.
- **Compute Optimized (c5, c6g):** High vCPU ratio for compute-intensive workloads, batch processing, and media encoding.
- **Memory Optimized (r5, x2gd):** High RAM ratio for in-memory databases (Redis, SAP HANA) and big data analytics.
- **Storage Optimized (i3, d3):** High sequential read/write IOPS for data warehousing and log processing.

4. EC2 User Data Script Example

Bootstrap an Apache web server automatically on launch:

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
```

```
echo 'Welcome to EC2 Cloud Server' > /var/www/html/index.html
```

KEY TAKEAWAYS & SUMMARY

- ✓ EC2 is the core IaaS (Infrastructure as a Service) offering in AWS.
- ✓ AMIs serve as blueprints for launching EC2 instances.
- ✓ Instance family selection depends on the bottleneck: CPU, Memory, Storage, or Graphics.